SQL Server Best Practices When Using Ipro® Software Q4 2019

Notices

Disclaimer

Information in this document, including URLs and other references, is subject to change without notice. Unless otherwise noted, any example companies, organizations, products, domain names, e-mail addresses, logos, people, locations, and events depicted herein are fictitious and no association with any real company, organization, product, domain name, e-mail address, logo, person, location, or event is intended or should be inferred. Complying with all applicable copyright laws is the responsibility of the user. Without limiting the rights under copyright, no part of this document may be reproduced, stored in or introduced into a retrieval system, or transmitted in any form or by any means (electronic, mechanical, photocopying, recording, or otherwise), or for any purpose, without the express written permission of Ipro Tech, LLC.
Ipro Tech, LLC may have patents, patent applications, trademarks, copyrights, or other intellectual property rights covering subject matter in this document. Except as expressly provided in any written license agreement from Ipro Tech, LLC, the furnishing of this document does not give the reader any license to these patents, trademarks, copyrights, or other intellectual property. Every precaution has been taken in the preparation of this document; however,
Ipro Tech does not assume responsibility for errors or omissions.

Trademarks

Ipro®, the Ipro logo, and all our product names are registered trademarks or trademarks of Ipro Tech, LLC. Please see http://iprotech.com/trademarks for details. Adobe is a registered trademark of Adobe Systems Incorporated in the United States and/or other countries. dtSearch is a registered trademark of dtSearch Corp. LEADTOOLS is a registered trademark of LEAD Technologies, Inc. Microsoft, Windows, and other Microsoft product names are either registered trademarks or trademarks of Microsoft Corporation in the United States and/or other countries. Oracle is a registered trademark of Oracle Corporation. Other products and company names mentioned herein may be trademarks of their respective companies and are used for editorial purposes and to the benefit of the trademark owner with no intention of infringing upon that trademark.
Some Ipro products use the Oracle® Outside In Viewer and TIF6 filter. The software is based in part on the work of the Independent JPEG Group.
Outside In Viewer © 1991, 2015 Oracle.

Copyright

© 2018 Ipro Tech, LLC All rights reserved. June 26, 2018
Published in the United States of America

Ipro Tech, LLC
1700 N. Desert Drive - Suite 101 Tempe, AZ 85281 USA

Contents

Microsoft SQL Server Best Practices 4
Installation 4
Security 4
Server Configuration 4
Data and Log File Management 5
Data and Log File Management (Continued) 6
Back Up/Restore Plan 6
Maintenance 6
Alerts Monitoring 7
Appendix 7

Microsoft SQL Server Best Practices

Installation

It is recommended that you use the most current:

- Hardware drivers
- Ipro-supported Operating System (*OS*) version and service pack
- Ipro-supported SQL Server version with the latest service pack and hot fixes

Note: Consult with Ipro Support or review the System Requirements documentation for all components to be installed and to confirm supported OS and SQL Server versions. Please check with Ipro Support if you wish to install additional components or other third-party applications.

Security

In order for Ipro products to be able to create and modify cases, a SQL authenticated account must be created with the following permissions:

- bulkadmin
- dbcreator
- public

When using multiple SQL servers:

- Linked Server objects must be created on each server and connected to each other server in the environment
- Use the same SQL username and password for all servers hosting the same instance

Note: To provide the greatest security for your environment, do not use the SA account for Ipro's Software.

Server Configuration

It is recommended that you:

- Do not run any other software or applications on the SQL Server
- Do not run any Antivirus locally
- Use 64-bit hardware and software (OS and SQL)
- Configure maximum server memory and leave 4GB for the OS

Data and Log File Management

- Build partitions with Industry Best Practice settings:
  - Partition Offset – All partitions regardless of purpose should be checked for partition alignment. 1024kb is recommended for all partitions, and is the Windows default.
  - Partition Cluster Size – All disks holding SQL data files (MDF, LDF, and TempDB) should use 64KB cluster size, which is different than the Windows default of 4KB.

Note: Storage device selection may dictate alternate settings for Partition Offset and Cluster Size. Consult with your storage vendor for recommendations on your specific environment.

- Specify a fixed size for log file growth, rather than percentage
- Pre-size data and log files to minimize autogrowth events
- Manage file growth manually (As a backup, leave autogrow on and set to a fixed size).
- Enable instant file initialization.
- Set MAXDOP accordingly
  - Max Degree of Parallelism – Typically set to the number of logical processors, not to exceed 8. We recommend 4 for most environments. For more information, please refer to https://support.microsoft.com/en-us/kb/2806535
  - Increase the Cost Threshold of Parallelism to 25 - 50
- Avoid putting any database files on the OS drive
- MDF data files should be on their own dedicated disks/spindles.
  - Store on fastest storage for read operations RAID 5-15K RPM
- LDF log files should be on their own dedicated disks/spindles.
  - Store on fastest storage for write operations RAID 10-15K RPM
- Tempdb system database should be on its own dedicated disks/spindles.
  - Store on fastest storage available RAID 10-15K RPM, Solid State Disk (*SSD*) if possible
  - Set initial size to the file's maximum size +10% undertypical load
  - Add multiple tempdb files as needed. Use at least one file for every four processors and up to one file per processor.
  - Keep the file size identical between each file.

Data and Log File Management (Continued)

- SQL system files (binaries) should be on their own dedicated disks/spindles.
  - Store on fastest storage for read operations RAID 5-15K RPM
- Validate volume performance with diskspd prior to installing SQL.
- Use full recovery mode on active review databases.
- Do NOT shrink database files without consulting with Ipro Support, as this is a very intensive resource-heavy process which can cause massive amounts of logical scan fragmentation and lead to poor performance.
- Run sp_delete_backuphistory to maintain space and backup performance (MSDB grows over time).

Back Up/Restore Plan

- On a daily basis, run a full back up on all system and production user databases.
- Back up transaction logs at least hourly (depending on activity and recovery point objective).
- Periodically test backups to verify they can be restored.
- Back up to a separate location from the data and log drives.
- DO NOT store backups long-term on the SQL Server.
- Maintain secure, off-site backups to mitigate physical threats.
- Create a Disaster Recovery Plan (DRP) that outlines step-by-step procedures to recover SQL Servers. Practice the DRP to verify it is complete (System owner should be familiar with this process).

Maintenance

- Rebuild indexes regularly and when index fragmentation is >30%.
- Reorganize indexes when fragmentation is 5%–30%.
- Use sys.dm_db_index_phsical_stats to identify fragmentation.
- Check OS, SQL Server, and error logs for any warnings, errors, or abnormal events.
- Schedule integrity checks at least weekly.
- Ensure you are keeping up with the latest service pack and hot fixes from Microsoft.

Alerts Monitoring	It is recommended that you: • Create a SQL Server event alert for all events with a severity of 19 (fatal) or higher • Create and configure Database Mail to ensure alerts are emailed to the party responsible for monitoring SQL Server health. • Consider using third party monitoring tools to provide real-time alerts of critical SQL Server resources, especially low disk space and low RAM (<25% of Operating System reserved memory free)

Appendix

Reference	Link
Sample Maintenance Scripts	http://ola.hallengren.com http://yoursqldba.grics.ca/index_en.shtml
Reference Materials	http://www.brentozar.com/blog/ https://docs.microsoft.com/en-us/previous-versions/sql/sql-server-2008/dd758814(v=sql.100) https://www.brentozar.com/archive/2015/09/getting- started-with-diskspd/